Metadynamics

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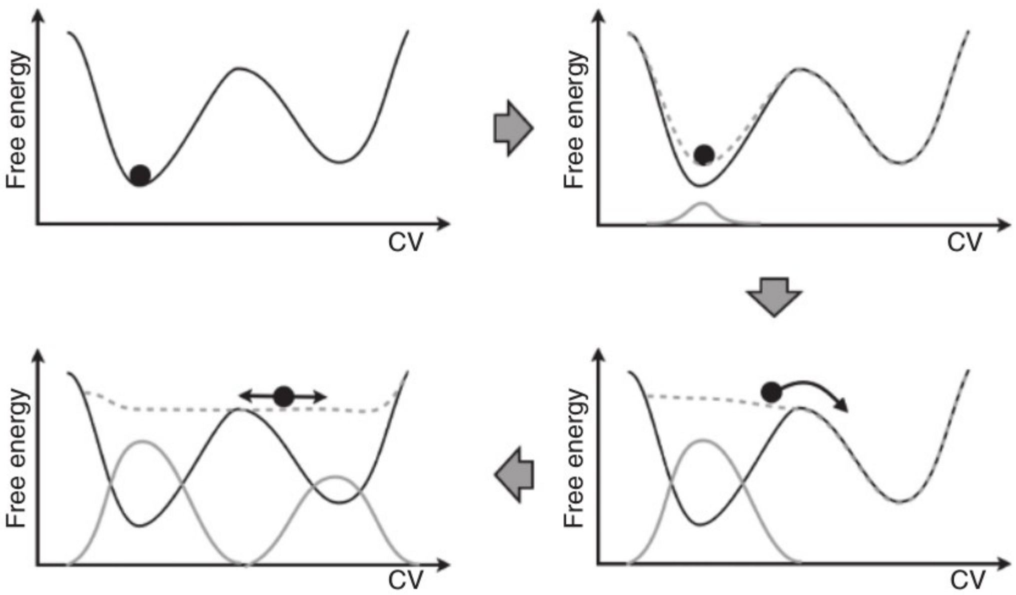
Metadynamics



A sketch of the process of metadynamics.

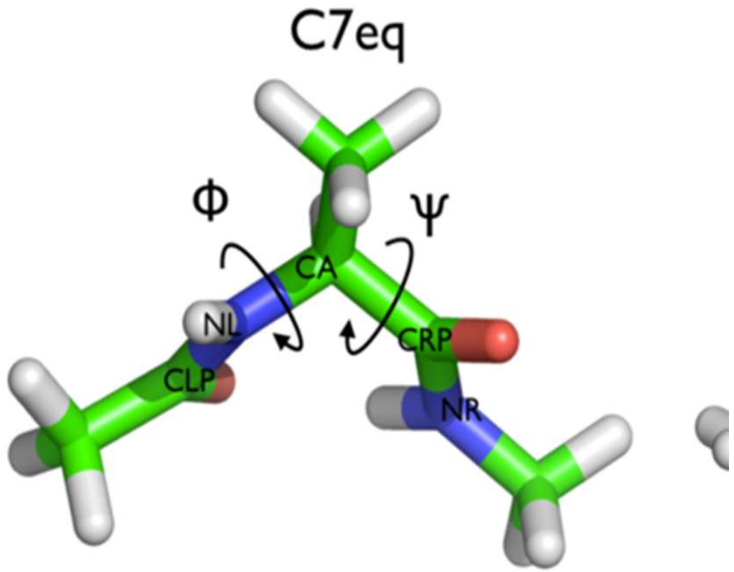
First the system evolves according to a normal dynamics, then a Gaussian potential is deposited (solid gray line). This lifts the system and modifies the free-energy landscape (dashed gray line) in which the dynamics evolves. After a while, the sum of Gaussian potentials fills up the first metastable state and the system moves into the second metastable basin. After this the second metastable basin is filled, at this point, the system evolves in a flat landscape. **The summation of the deposited bias (solid gray profile) provides a first rough negative estimate of the free-energy profile**.

https://chemistry-europe.onlinelibrary.wiley.com/doi/full/10.1002/cphc.200500427

Metadynamics

我们基于plumed官方教学学习基本的操作，创建一个dialaA.pdb，内容如下，我们将要在phi角上增加偏置，增强该角度的采样

```
TITLE     alanine dipeptide in vacuum
MODEL        1
ATOM      1 1HH3 ACE     1      20.900  12.650  12.200  1.00  0.00           H
ATOM      2  CH3 ACE     1      20.000  12.240  11.740  1.00  0.00           C
ATOM      3 2HH3 ACE     1      19.440  13.160  11.580  1.00  0.00           H
ATOM      4 3HH3 ACE     1      19.470  11.550  12.400  1.00  0.00           H
ATOM      5  C   ACE     1      20.380  11.520  10.460  1.00  0.00           C
ATOM      6  O   ACE     1      21.560  11.390  10.140  1.00  0.00           O
ATOM      7  N   ALA     2      19.380  11.100   9.680  1.00  0.00           N
ATOM      8  H   ALA     2      18.430  11.250  10.010  1.00  0.00           H
ATOM      9  CA  ALA     2      19.520  10.590   8.330  1.00  0.00           C
ATOM     10  HA  ALA     2      20.290   9.830   8.200  1.00  0.00           H
ATOM     11  CB  ALA     2      18.160   9.920   8.100  1.00  0.00           C
ATOM     12  HB1 ALA     2      17.990   9.210   8.910  1.00  0.00           H
ATOM     13  HB2 ALA     2      17.400  10.700   8.010  1.00  0.00           H
ATOM     14  HB3 ALA     2      18.200   9.370   7.160  1.00  0.00           H
ATOM     15  C   ALA     2      19.860  11.700   7.350  1.00  0.00           C
ATOM     16  O   ALA     2      19.780  12.880   7.680  1.00  0.00           O
ATOM     17  N   NME     3      20.310  11.280   6.170  1.00  0.00           N
ATOM     18  H   NME     3      20.520  10.300   6.130  1.00  0.00           H
ATOM     19  CH3 NME     3      20.440  12.050   4.940  1.00  0.00           C
ATOM     20 1HH3 NME     3      19.440  12.240   4.540  1.00  0.00           H
ATOM     21 2HH3 NME     3      21.010  12.970   5.040  1.00  0.00           H
ATOM     22 3HH3 NME     3      20.890  11.430   4.170  1.00  0.00           H
TER
ENDMDL
```



https://www.plumed-tutorials.org/lessons/21/004/data/INSTRUCTIONS.html

Run a classic MD first

首先使用以下脚本运行一个cMD，方便后续与metad对比：

- 注意echo命令的使用，它在这里的作用是什么呢？
- 最后一行gmx mdrun命令使用nsteps做了什么呢？

使用的mdp文件完全按照之前蛋白质模拟（因为我们模拟的二肽也是蛋白质），注意我为了后续作图效果好，把md.mdp里xtc格式的轨迹输出频率变成10步输出一次（这么做会极大拖慢模拟速度，所以不建议这么做）。

```
echo -e "6\n1" | gmx pdb2gmx -f dialaA.pdb -o dialaA.gro -p dialaA.top
gmx editconf -f dialaA.gro -o dialaA_box.gro -c -d 1.2 -bt cubic
gmx solvate -cp dialaA_box.gro -cs spc216.gro -o solv.gro -p dialaA.top
gmx grompp -f ions.mdp -c solv.gro -p dialaA.top -o ions.tpr
echo -e "13\n" | gmx genion -s ions.tpr -o ions.gro -p dialaA.top -pname NA -nname CL -conc 0.15 -neutral
gmx grompp -f em.mdp -c ions.gro -p dialaA.top -o em.tpr
gmx mdrun -v -deffnm em
gmx grompp -f nvt.mdp -c em.gro -p dialaA.top -r em.gro -o nvt.tpr
gmx mdrun -v -deffnm nvt
gmx grompp -f npt.mdp -c nvt.gro -p dialaA.top -r nvt.gro -o npt.tpr
gmx mdrun -v -deffnm npt
gmx grompp -f md.mdp -c npt.gro -p dialaA.top -r npt.gro -o md.tpr
gmx mdrun -s md.tpr -nsteps 10000000 -x cMD.xtc
```

Calculate dihedral values with plumed

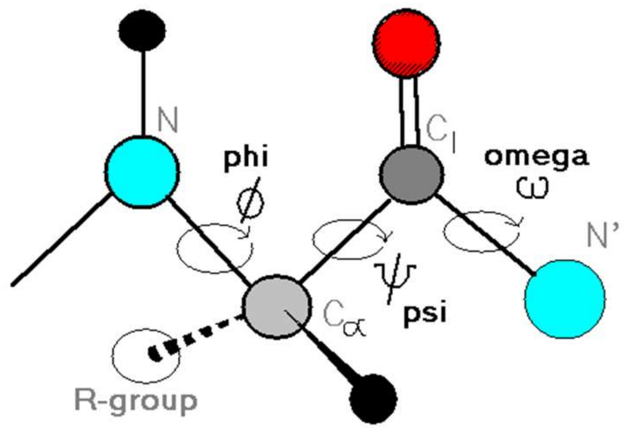
cal_cv.dat

```
# Activate MOLINFO functionalities
MOLINFO STRUCTURE=dialaA.pdb
# Compute the backbone dihedral angle phi, defined by atoms C-N-CA-C
# you should use MOLINFO shortcuts
phi: TORSION ATOMS=5,7,9,15
# Compute the backbone dihedral angle psi, defined by atoms N-CA-C-N
# here also you should use MOLINFO shortcuts
psi: TORSION ATOMS=7,9,15,17
# Print the two collective variables on COLVAR file every step
PRINT ARG=phi,psi FILE=cv_cMD STRIDE=1
```

使用左侧的控制文件可以计算cMD轨迹phi与psi随时间变化的情况；

阅读及猜测控制文件每一行的意义？

```
plumed driver --plumed cal_cv.dat --mf_xtc cMD.xtc
```



cv_file

ta > cv_file

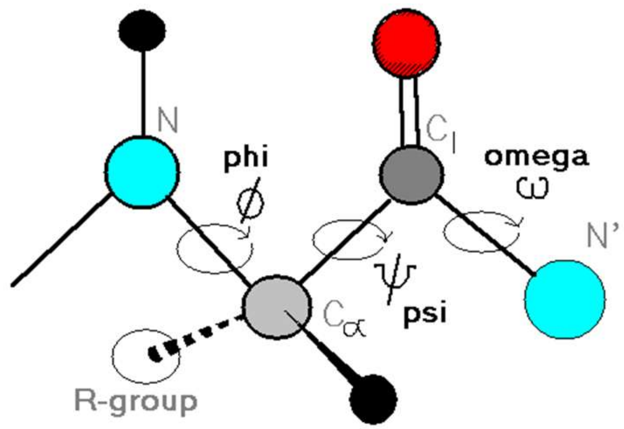
```
#! FIELDS time phi psi
#! SET min_phi -pi
#! SET max_phi pi
#! SET min_psi -pi
#! SET max_psi pi
 0.000000 -1.521112 3.031219
 1.000000 -2.244389 2.717322
 2.000000 -1.299374 2.129855
 3.000000 -2.543502 2.536684
 4.000000 -1.531260 -3.138126
 5.000000 -1.191381 -0.211075
 6.000000 -0.957034 -0.599277
 7.000000 -1.314121 -0.173140
 8.000000 -2.287308 2.793289
 9.000000 -1.208228 2.295788
 10.000000 -2.017675 2.468813
 11.000000 -1.550348 0.151231
 12.000000 -1.282605 -0.325253
 13.000000 -1.647121 -0.586915
 14.000000 -0.846082 2.723643
 15.000000 -1.071819 -0.249655
 16.000000 -1.325836 -0.163993
 17.000000 -2.100200 -0.108536
 18.000000 -0.922194 -0.570552
 19.000000 -1.736894 3.104432
 20.000000 -2.443694 -0.253409
```

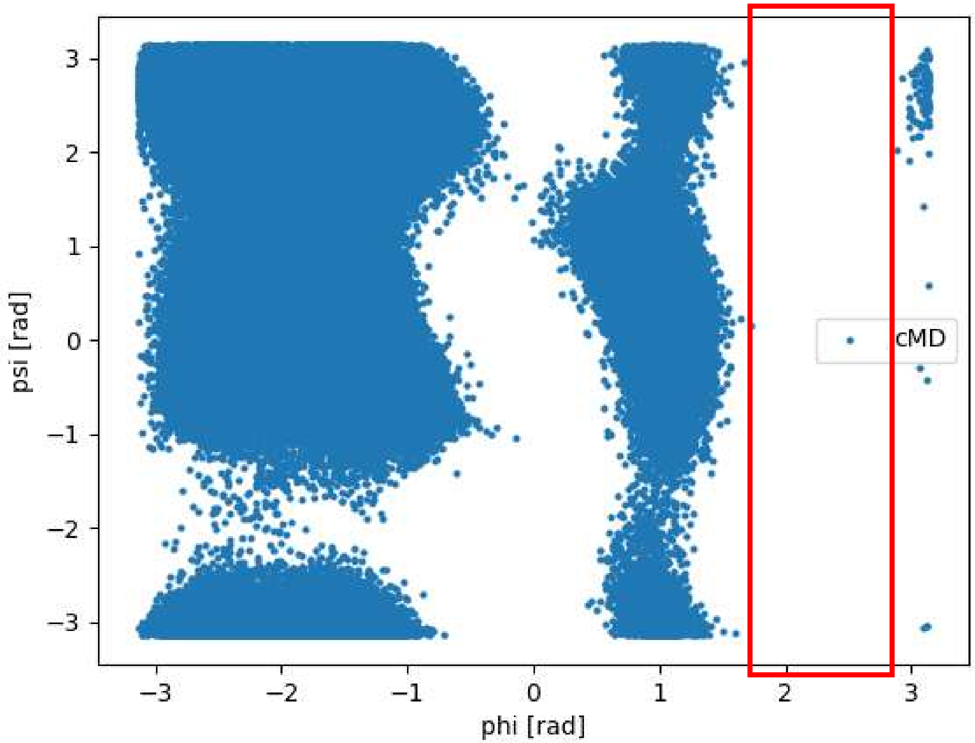
Calculate dihedral values with plumed

cal_cv.dat

```
# Activate MOLINFO functionalities
MOLINFO STRUCTURE=dialaA.pdb
# Compute the backbone dihedral angle phi, defined by atoms C-N-CA-C
# you should use MOLINFO shortcuts
phi: TORSION ATOMS=5,7,9,15
# Compute the backbone dihedral angle psi, defined by atoms N-CA-C-N
# here also you should use MOLINFO shortcuts
psi: TORSION ATOMS=7,9,15,17
# Print the two collective variables on COLVAR file every step
PRINT ARG=phi,psi FILE=cv_cMD STRIDE=1
```

```
plumed driver --plumed cal_cv.dat --mf_xtc cMD.xtc
```





观察phi角的分布，明显能看到cMD在某些区域没能成功采样

Run metadynamics

plumed_md.dat

```
# Activate MOLINFO functionalities
MOLINFO STRUCTURE=dialaA.pdb
# Compute the backbone dihedral angle phi, defined by atoms C-N-CA-C
# you should use MOLINFO shortcuts
phi: TORSION ATOMS=@phi-2
# Compute the backbone dihedral angle psi, defined by atoms N-CA-C-N
# here also you should use MOLINFO shortcuts
psi: TORSION ATOMS=@psi-2
# Activate well-tempered metadynamics in phi
metad: METAD ARG=phi ...
# Deposit a Gaussian every 500-time steps, with initial height
# equal to 1.2 kJ/mol and bias factor equal to 8
  PACE=500 HEIGHT=1.2 BIASFACTOR=8
# Gaussian width (sigma) should be chosen based on the CV fluctuations
in unbiased run
# try 1/2 or 1/3 of the estimated fluctuations
  SIGMA=0.3
# Gaussians will be written to file and also stored on grid
  FILE=HILLS GRID_MIN=-pi GRID_MAX=pi
...
# Print both collective variables on COLVAR file every 10 steps
PRINT ARG=phi,psi FILE=COLVAR STRIDE=10
```

我们使用metad对phi进行增强采样

使用dialaA.pdb作为参考结构以进行后续的原子选择

基于dialaA.pdb选择组成第二个残基phi角的原子

METAD关键字，设置将偏置加在CV phi上的metadynamics

每500步增加一次偏置；设置height和biasfactor

设置SIGMA

偏置写入HILLS文件，phi的采样范围在-pi到pi之间

每10步将phi与psi的值写入COLVAR文件

```
gmx mdrun -s md.tpr -nsteps 10000000 -plumed plumed_md.dat -x metad.xtc
```

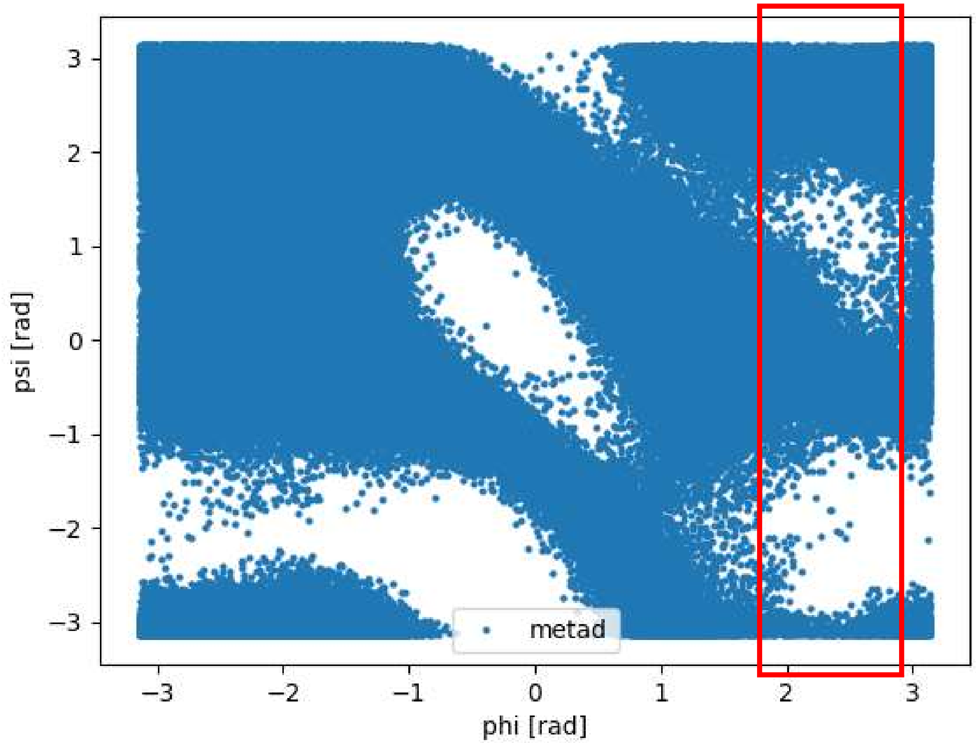
Run metadynamics

cal_cv2.dat

```
# Activate MOLINFO functionalities
MOLINFO STRUCTURE=dialaA.pdb
# Compute the backbone dihedral angle phi, defined by atoms C-N-CA-C
# you should use MOLINFO shortcuts
phi: TORSION ATOMS=5,7,9,15
# Compute the backbone dihedral angle psi, defined by atoms N-CA-C-N
# here also you should use MOLINFO shortcuts
psi: TORSION ATOMS=7,9,15,17
# Print the two collective variables on COLVAR file every step
PRINT ARG=phi,psi FILE=cv_metad STRIDE=1
```

```
plumed driver --plumed cal_cv2.dat --mf_xtc metad.xtc
```

虽然我这也加了这一步，但其实这样得到的cv_metad和上一页得到的COLVAR应该是一样的

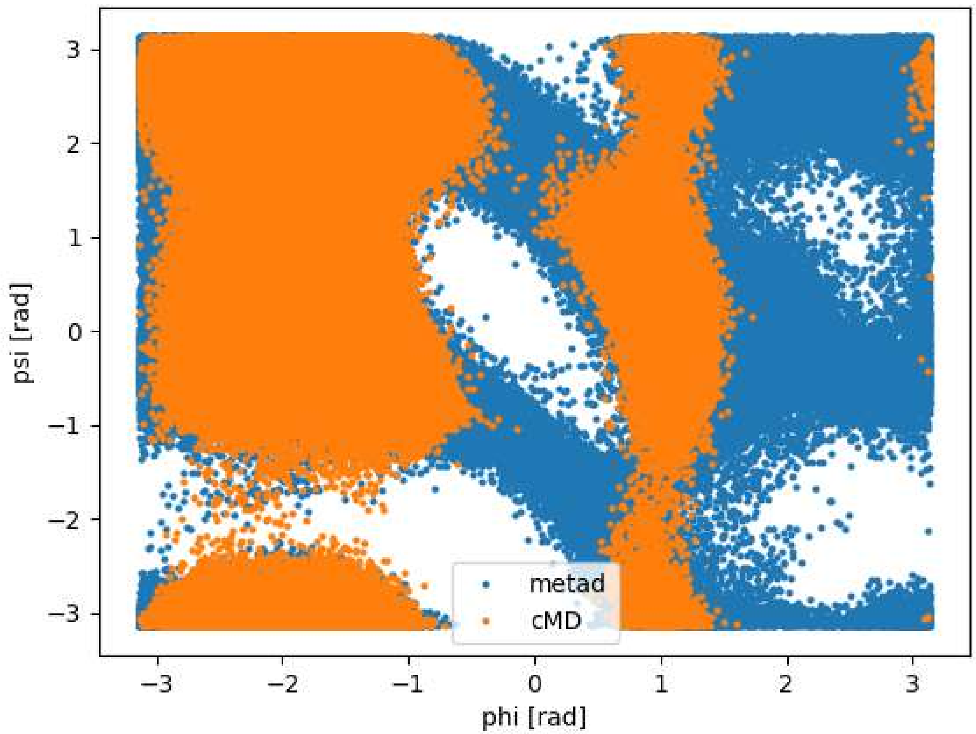


我们注意到metad对phi的采样确实强了，cMD没采到的phi角区域通过metad成功获取了

Comparing the sampling efficiency of cMD and metad

```
import plumed
import matplotlib.pyplot as plt

# import PLUMED COLVAR files into pandas datasets
data_metad =plumed.read_as_pandas("COLVAR")
data_cMD=plumed.read_as_pandas("cv_cMD")
# plot phi vs psi
plt.plot(data_metad ["phi"], data_metad ["psi"],'o', label="metad")
plt.plot(data_cMD["phi"], data_cMD["psi"],'o', label="cMD")
# x-y axis labels
plt.xlabel("phi [rad]")
plt.ylabel("psi [rad]")
plt.legend()
# calculate Gaussian sigma
print("Std phi metad:",np.std(data_metad["phi"]))
print("Std phi cMD:",np.std(data_cMD["phi"]))
plt.savefig('cv.png')
```



Converge

plumed sum_hills --hills HILLS --stride 200 --mintozero

```
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   opening file HILLS
PLUMED:   done with this chunk: now with 200 kernels
PLUMED:
PLUMED:   Writing full grid on file fes_0.dat
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   done with this chunk: now with 400 kernels
PLUMED:
PLUMED:   Writing full grid on file fes_1.dat
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   done with this chunk: now with 600 kernels
PLUMED:
PLUMED:   Writing full grid on file fes_2.dat
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   done with this chunk: now with 800 kernels
PLUMED:
PLUMED:   Writing full grid on file fes_3.dat
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   done with this chunk: now with 1000 kernels
PLUMED:
PLUMED:   Writing full grid on file fes_4.dat
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   done with this chunk: now with 1200 kernels
PLUMED:
PLUMED:   Writing full grid on file fes_5.dat
PLUMED:   reading hills:
PLUMED:   doing serialread
PLUMED:   done with this chunk: now with 1400 kernels
```

sum_hills命令会根据HILLS里写入的偏置项参数复原采样CV的势能面；
记得吗，最终使用的所有偏置项之和的相反数，它的形状就是CV的势能面加一个常数。

注意到fes_0.dat数据来自前200个kernel（就是加了前200个偏置）；fes_1.dat数据来自前400个；

也就是说，这个命令会根据stride增加用于计算自由能面的轨迹长度。

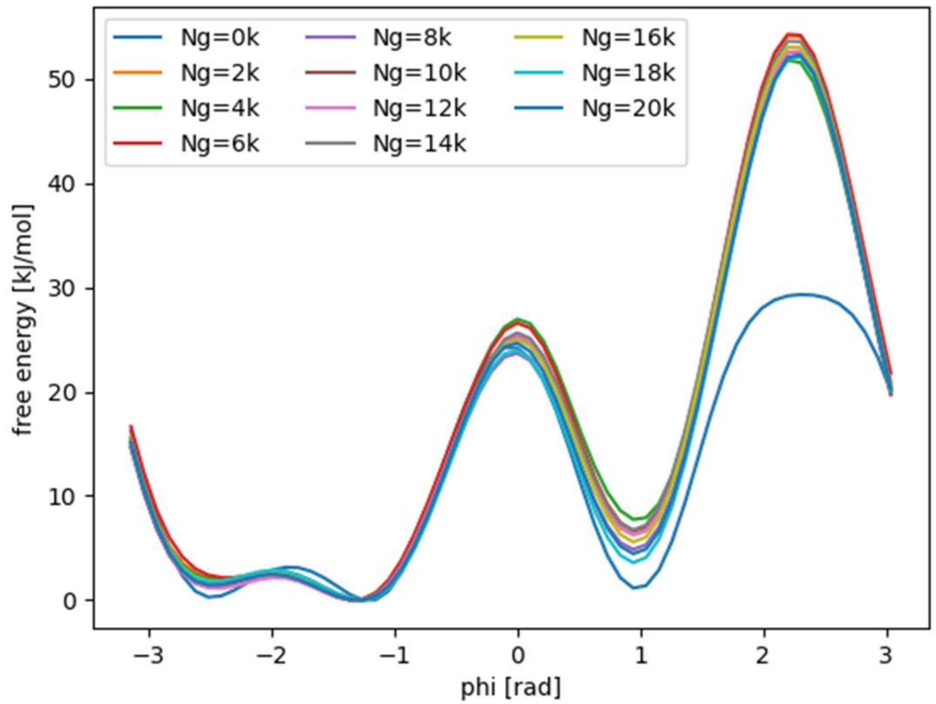
Converge

```
# plot free energy as a function of simulation time

for i in range(0,101,10):
    # import fes file into pandas dataset
   data=plumed.read_as_pandas("fes_"+str(i)+".dat")
   # plot fes
   plt.plot(data["phi"],data["file.free"], label="Ng="+str(int(2*i/10))+"k")

# labels

plt.xlabel("phi [rad]")
plt.ylabel("free energy [kJ/mol]")
plt.legend(ncol=3)
```

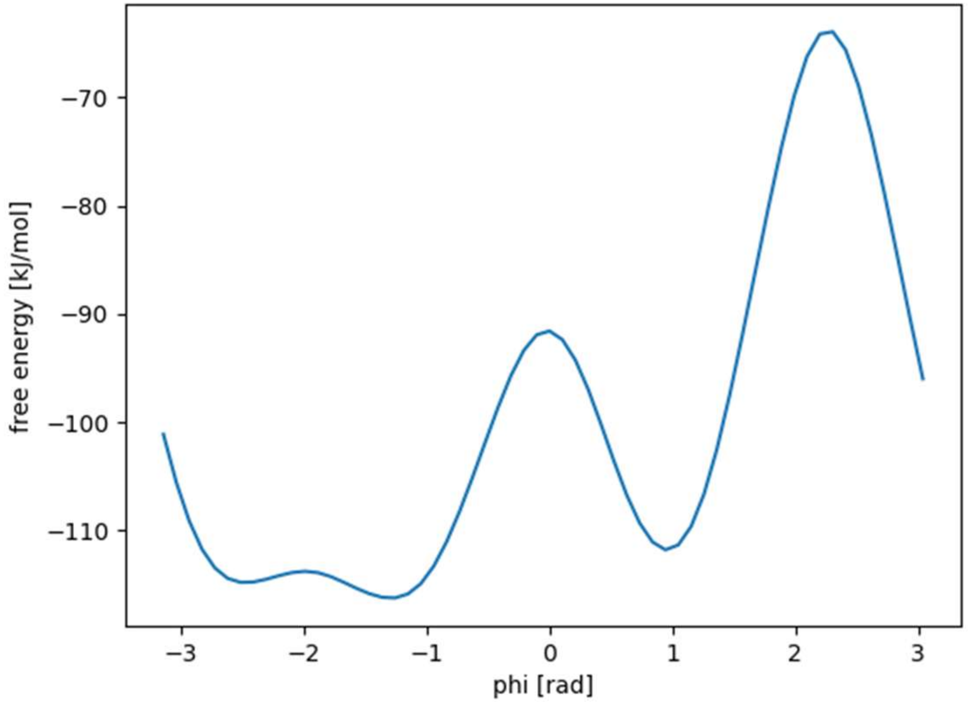


当添加更多的轨迹不影响自由能面形状的时候，说明模拟收敛了

Plot the free energy surface along phi

```
plumed sum_hills --hills HILLS
```

```
data=plumed.read_as_pandas("fes.dat")
 # plot fes
plt.plot(data["phi"],data["file.free"])
# labels
plt.xlabel("phi [rad]")
plt.ylabel("free energy [kJ/mol]")
plt.savefig('sumhill.png')
```



确定轨迹收敛以后可以直接用轨迹全长绘制势能面

Reweighting

https://www.plumed.org/doc-v2.9/user-doc/html/master-_i_s_d_d-2.html

尝试根据官方教程学习做reweighting？